Indian Institute of Technology Kanpur
CS771 Introduction to Machine Learning

Instructor: Purushottam Kar
Date: March 4, 2026
Total: 50 marks

MINOR ASSIGNMENT
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1 What should I submit, where should I submit and by when?

Your submission for this minor assignment will be one PDF (.pdf) file only. Instructions on how to prepare and submit this file are given below.

Assignment Package:
`https://www.cse.iitk.ac.in/users/purushot/courses/ml/2025-26-w/material/assignments/minor1.zip`
Deadline for all submissions: March 18, 2026, 9:59PM IST
Code Submission: ***None needed – do not submit any code (ipynb, py, etc) files for this minor assignment***
Report Submission: on Gradescope
There is no provision for "late submission" for this minor assignment.

1.1 How to submit the PDF report file

1. The PDF file must be submitted using Gradescope in the *group submission mode*.
2. Note that unregistered auditors cannot make submissions to this minor assignment.
3. Make only one submission per assignment group on Gradescope, not one submission per student. Gradescope allows you to submit in groups - please use this feature to make a group submission.
4. Link all group members in your group submission. If you miss out on a group member while submitting, Gradescope will think that person never submitted anything.
5. You may overwrite your group's submission as many times as you want before the deadline (submitting again on Gradescope simply overwrites the old submission).
6. Do not submit Microsoft Word or text files. Prepare your report in PDF using the style file we have provided (instructions on formatting given later).

Problem 1.1 (Testing the Limits). Melbob wishes to understand the effects of various SVM hyperparameters using a 2D problem where one can visualize the data as well as the output.

Your Data. We have provided you with files to generate 2D synthetic data with varying levels of challenge and plot the data and output of any classifier learnt on that data. The data can be generated using the `genChallengeData` function and takes two arguments

1. `level` should be a non-negative number strictly greater than zero
2. `n` should be a non-negative integer as it decides the number of points

(50 marks)

Your Task. The following enumerates 7 parts to the question. All parts need to be answered in the PDF file containing your report. **Do not submit any code files for this minor assignment**. Thus, submit only a single PDF file as your submission.

1. Inspect the function `genChallengeData` in the file `genSyntheticData.py`. Will data generated by this function be linearly separable for all values of `level` and `n`? Answer **Yes** or **No** along with either a mathematical proof or else a counterexample. (10 marks)

2. For small challenge levels (`level < 0.5`), experimentally find out the smallest value for the hyperparameters `myC, max_iter` and largest value of the hyperparameter `myTol` for which the SVM solver still yields 100% accuracy. Demonstrate your work by showing accuracy for various values of these hyperparameters. You must explore sufficiently many (at least 10) combination of these hyperparameters to get full marks for this part. Plot the figure showing the decision boundary using the `shade2D` and `plot2D` as shown in the Jupyter notebook for the best values of the hyperparameters you have found. (10 marks)

3. For large challenge levels (`level > 50`), experimentally find out the smallest value for the hyperparameters `myC, max_iter` and largest value of the hyperparameter `myTol` for which the SVM solver still yields 100% accuracy. Demonstrate your work by showing accuracy for various values of these hyperparameters. You must explore sufficiently many (at least 10) combination of these hyperparameters to get full marks for this part. Plot the figure showing the decision boundary using the `shade2D` and `plot2D` as shown in the Jupyter notebook for the best values of the hyperparameters you have found. (10 marks)

4. Find out the effect of at least three hyperparameters out of
   `myPenalty, myC, myLoss, max_iter, myTol`
   on accuracy and training time when used on medium challenge level (e.g. `myLevel = 5`). For each hyperparameter, you must explore at least 4 values of that hyperparameter. Refer to the sklearn API for valid values of these hyperparameters `https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html`. Note that certain combinations e.g. `penalty='l1'` and `loss='hinge'` are not supported. (10 marks)

5. How does the number of training points `n` affect train time and accuracy? Explore at least 4 values of `n` (small e.g. 10 all the way to large e.g. 1000) on high challenge level (`myLevel > 50`). (10 marks)

Using Internet Resources. You are allowed to refer to textbooks, internet sources, research papers to find out more about this problem and for specific derivations e.g. the arbiter-PUF problem. However, if you do use any such resource, cite it in your PDF file. There is no penalty for using external resources with attribution but claiming someone else's work (e.g. a book or a research paper) as one's own work without crediting the original author will attract penalties.

Restrictions on Code Usage. You should not use any library other than what is already imported in the Jupyter notebook in the assignment package (you do not need to anyway). Since this minor assignment does not require any code submission, we will trust you on this one. However, you may use fancier libraries like seaborn or plotly to plot graphs etc.

2 How to Prepare the PDF File

Use the following style file to prepare your report. The ZIP file contains an example file and instructions.

https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip

You must use the following command in the preamble

```
\usepackage[preprint]{neurips_2025}
```

instead of `\usepackage[preprint]{neurips_2025}`. Use proper LaTeX commands to neatly typeset your responses to the various parts of the problem. Use neat math expressions to typeset your derivations. Remember that all parts of the question need to be answered in the PDF file. All plots must be generated electronically - hand-drawn plots are unacceptable. All plots must have axes titles and a legend indicating what are the plotted quantities. Insert the plot into the PDF file using LaTeX `\includegraphics` commands.